



Karnataka State Police
Government of Karnataka

Powered by **H2S**

0atathon

2026

Technology partner



Z O H O

Team Details

- a. **Team name:** Daku Mangal Singh
- b. **Team leader name:** Roodraksh Agrawal
- c. **Team size:** 5
- d. **Problem Statement:**

PROBLEM STATEMENT

AI-powered crime investigation and decision support for the State Crime Records Bureau. Fragmented FIRs make it hard to connect suspects, vehicles and crime patterns, or track critical filing deadlines. Build a secure bilingual assistant that unifies records, reveals cross-case links and supports timely, evidence-based action while preserving officer judgement.



Brief about the solution

SCRB Sahayak is a web workspace that sits on the station's case records and does the reading, cross-referencing, and drafting an investigating officer would otherwise do by hand.

- **Ask in plain language**

The officer types or speaks a question in English or Kannada. The copilot answers from the records and shows which FIRs it used.

- **See the connections**

Every case, person, and vehicle becomes a node in one graph. It detects crime rings and ranks who holds the network together.

- **Never miss a deadline**

It tracks the BNSS 187(3) charge-sheet clock for every open case and warns before an accused becomes eligible for default bail.

- **Draft the paperwork**

From an FIR scan it fills the case record, suggests the likely IPC and BNS sections, and drafts a Section 173 charge sheet the officer edits.



Opportunities

- a. How different is it from any of the other existing ideas?
- b. How will it be able to solve the problem?
- c. USP of the proposed solution

Different from existing tools

Most tools stop at a searchable case list. This one runs graph analysis over the records and reads scanned FIRs on device.
It tracks the statutory default-bail clock, which case-management software usually ignores, and works offline with no paid map key.

How it solves the problem

Links a suspect, plate, or method across every FIR in the officer's jurisdiction in seconds.
Flags cases about to breach the 60 or 90 day filing window, and cuts FIR entry and charge-sheet drafting to a review step.

USP of the proposed solution

Broker detection names the person whose removal fragments a network, not just the busiest one.
Every answer cites its FIR sources into an immutable audit log, and access mirrors the real Karnataka Police rank hierarchy.



List of features offered by the solution

- **Bilingual copilot**
English and Kannada Q&A grounded in case records, with a 15-tool calling loop.
- **Entity network graph**
Cases, people, and vehicles in one canvas. Ring detection, hubs, brokers.
- **Section predictor**
Suggests likely IPC and matching BNS sections, with the evidence to secure.
- **FIR intake with OCR**
Reads scanned FIRs in English and Kannada as a background job, extracts fields.
- **Explainable and audited**
Answers cite their FIR sources; every query, upload, and export is logged.
- **Officer onboarding**
Police IT invites officers, emails temporary passwords, and handles resets.
- **Voice in and out**
Speech recognition for questions and spoken answers, aware of Kannada.
- **Default-bail shield**
Tracks the BNSS 187(3) clock per case and tiers them overdue, urgent, watch.
- **Charge-sheet drafting**
Generates a Section 173 CrPC report the officer edits and saves.
- **Hotspots and warnings**
A Leaflet map plus a risk engine that recomputes when a new case is filed.
- **Rank hierarchy access**
15 ranks. Data scope scales from one station up to the whole state.
- **Quick-ask pill**
A floating assistant that answers about the current page without leaving it.



Process flow diagram or Use-case diagram

1 Upload FIR

Officer drops a scanned FIR (image or PDF).

2 OCR read

Tesseract reads it in English and Kannada.

3 Extract fields

The model pulls accused, victim, date, sections.

4 Match leads

Identity and method matches against prior cases.

5 Confirm and save

Officer edits, confirms leads, saves the case.

6 Intelligence

Case joins the map, network, and deadline board.

WHO USES IT

Constable and SI:

File FIRs, ask the copilot, confirm leads.

Inspector and SHO:

Run the network, watch deadlines, draft charge sheets.

SP and gazetted ranks:

District to state view, analytics, oversight.

Police IT:

Onboard officers, manage access and password resets.

Each officer sees only their level: a constable sees one station, an SP a district, senior ranks up to the whole state.



Wireframes/Mock diagrams of the proposed solution (optional)

The workspace is one screen: a rank-scoped menu on the left, the copilot thread in the center, and a live status rail on the right.

- **Left navigation**

An officer only sees the sections their command level allows.

- **Center thread**

The copilot conversation. Answers carry source chips and a spoken-playback control.

- **Floating pill**

Page-aware quick questions that answer in place and are not saved to history.

- **Right rail**

A live needs-attention block: lapsed clocks and leads waiting on a decision.

- **Records pages**

Cases, network, deadlines, hotspots, and audit each have their own screen.

Paste UI screenshots here if the template allows an image on this page.



Architecture diagram of the proposed solution

CLIENT

React 18, Vite, TypeScript. Tailwind, Framer Motion, Zustand. Recharts for charts, Leaflet for maps.

API (FastAPI, Python)

JWT auth with bcrypt and rank-scoped access. Copilot tool loop, analytics, network builder, hotspot warning engine. Background jobs for OCR intake, email for invitations.

DATA AND SERVICES

PostgreSQL via raw psycopg with an ltree rank hierarchy. OpenRouter LLM, Tesseract OCR, edge-TTS. OpenStreetMap and CARTO tiles, SMTP for mail.

Principle: The frontend never touches the database. All access goes through the API, which enforces rank-based scoping and writes an audit row for every read of case data.



Technologies to be used in the solution

Frontend

React 18, Vite, TypeScript, Tailwind CSS, Framer Motion, Zustand, Recharts, Leaflet with OpenStreetMap.

Backend

Python, FastAPI, psycpg (raw SQL), bcrypt, JWT, Pydantic, background tasks and SMTP email.

Data and AI

PostgreSQL with Itree, OpenRouter LLM, Tesseract OCR, PyMuPDF, edge-TTS and jsPDF.

Deployment & DevOps

Docker and Docker Compose for portable, reproducible deployment across local, data-centre and cloud environments.



List down Catalyst Services being used in the solution

The prototype runs on open, self-hostable services, so it works without a paid map key or a single cloud vendor. Each can move to a managed service, including Zoho Catalyst functions, datastore, and mail, for production.

OpenRouter

LLM gateway for the copilot, field extraction, and charge-sheet drafting.

PostgreSQL / Supabase

Case records, rank hierarchy, and audit log. Local Docker or managed Postgres.

OpenStreetMap + CARTO

Hotspot map tiles with no API key and no request cap.

Microsoft edge-TTS

Spoken answers, including a Kannada voice.

Tesseract (self-hosted)

Reads scanned FIRs in English and Kannada on the server.

SMTP

Sends officer invitations and password-reset credentials.



Estimated implementation cost (optional)

Indicative single-district pilot comparison

	SCRB SAHAYAK	CONVENTIONAL POLICE IT SETUP
Upfront setup	Rs. 0-15,000	Rs. 2-5 lakh
Monthly operations	Rs. 3,000-5,000	Rs. 40,000-75,000
Deployment time	1-2 days	4-8 weeks
Map, OCR & TTS	Included / self-hosted	Often paid add-ons
Portability	Docker: local, DC or cloud	Vendor-specific setup
Officer capability	Bilingual AI + linked insights	Search and manual review

Outcome: about 85-90% lower monthly cost, faster rollout, and richer investigation support.

Illustrative estimates; actual government procurement, infrastructure and usage costs may vary.



Snapshots of the prototype

Working screens from the running build. Paste your own screenshots beside each label before submitting.



Investigation Copilot

Bilingual chat with cited FIR sources and a needs-attention rail.



Statutory deadlines

Default-bail clock per case with charge-sheet drafting.



FIR intake with OCR

A scan read in English and Kannada, fields extracted.



Entity network

Cross-case graph with ring detection, hubs, and broker ranking.



Hotspot map

Crime clusters on a Leaflet map with risk ranking.



Police IT console

Officer directory, invitations, and password resets by rank.



Prototype Performance report/Benchmarking

68

backend tests passing

11x

faster analytics after tuning

eng+kan

bilingual OCR in one pass

15

ranks in the access hierarchy

API response time on a remote database (lower is better):

	before	after
Analytics	3.5 s	0.30 s
Hotspots	2.3 s	0.30 s
Network	1.9 s	0.35 s
Dashboard	1.8 s	0.30 s

Also:

- Independent queries run at the same time, not one after another.
- Short-lived caching absorbs tab switching without re-querying.
- Charge-sheet drafting produces a full Section 173 report.
- FIR OCR runs as background jobs, so navigating away keeps it.



Provide links to your:

1. **GitHub Public Repository**
2. **Demo Video Link (3 Minutes)**
3. **Deployed Link**

<https://github.com/Roodraksh12/KSPAi>

https://www.youtube.com/watch?v=_UJdQK79kRM

<https://kspai-zgymgiew.onslate.in/>

Demo logins (password demo1234):

KA-SP-9999 KA-INS-4471 KA-CON-1001 KA-IT-0001 (Police IT admin)



Additional Details/Future Development (if any)

- **Integrate with CCTNS**

Pull real FIR and case data from the national Crime and Criminal Tracking Network.

- **Smaller on-device model**

A faster model for routine questions to cut quick-ask latency and token cost.

- **Court-filing export**

One-click export of the charge sheet in the format the local court accepts.

- **Face and plate recognition**

Match CCTV faces and number plates against the case database, into the network graph.

- **More Indian languages**

Extend beyond Kannada to Hindi, Tamil, and Telugu for use across states.

- **Supervisor analytics**

District and range dashboards for senior ranks to spot slipping cases.



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THANK YOU